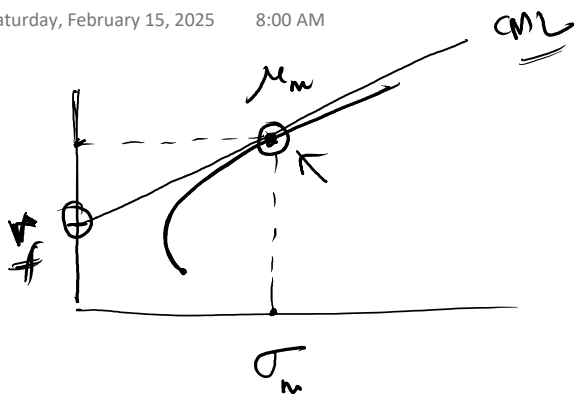




$$E[r_m] - r_f = \bar{\gamma} \sigma_m^2$$

$$U = \underline{\underline{E(r_p)}} - \frac{1}{2}(\underline{\underline{\gamma}}) \underline{\underline{V(r_p)}}$$

(mean-variance utility function)

$$V(r_p) = \underbrace{(w\sigma_m)^2}_{+ (1-w)^2 \sigma_m^2} + 2 \dots$$

$$U(w) = w\mu_m + (1-w)r_f - \frac{1}{2}\gamma \cdot (w\sigma_m)^2$$

$$\frac{dU}{dw} = 0$$

$$\mu_m - r_f - \gamma w \sigma_m^2 = 0$$

$$\Rightarrow w^* = \frac{\mu_m - r_f}{\gamma \sigma_m^2} = 1$$

Avg investor

$$\underline{\underline{E[r_m] - r_f}} = \underline{\underline{(\bar{\gamma}) \sigma_m^2}}$$

$$E[R_i] - r_f = \beta_1 E[R_m - r_f] + \beta_2 E[SMB] + \beta_3 E[HML]$$

$$\textcircled{+} \quad \underline{\underline{R_i - r_f}} = \underline{\underline{\alpha}} + \underline{\underline{\beta_1}} (R_m - r_f) + \underline{\underline{\beta_2}} (SMB) + \underline{\underline{\beta_3}} (HML) + \textcircled{\epsilon_i}$$